Plants, Part 1: Structure & Transpiration

Answer sheet · 31 marks total

Section A — Plant organs & meristem tissue

A1. Root: anchors the plant / absorbs water and mineral ions (1). Stem: supports the plant / holds leaves up to the light / transports substances (1). Leaf: main site of photosynthesis / gas exchange (1).

A2. Unspecialised (undifferentiated) cells that continuously divide by mitosis to produce new cells (1).

A3. At the tips of roots and shoots (root tip / shoot tip) (1).

A4. New cells produced by meristem tissue can differentiate into specialised cells (1), allowing the plant to grow new tissue/organs throughout its life, not just when young (1).

Section B — Xylem & phloem

B1. Xylem (1).

B2. Phloem (1).

B3. Xylem: one direction only, upwards, from roots to leaves (1). Phloem: both directions, up and down (1).

B4. No cytoplasm/being dead leaves a hollow tube, so water can flow through with little resistance (1); lignin strengthens the walls, helping xylem withstand the pressure of water movement and support the plant (1).

Section C — Transpiration

C1. The loss of water vapour (1) from a plant's leaves, mainly through the stomata, by evaporation and diffusion (1).

C2. Absorbed from the soil by root hair cells, by osmosis (1); moves across the root into the xylem (1); travels up the xylem through the stem to the leaves (1); evaporates from cells inside the leaf and diffuses out through the stomata (1).

C3. Stomata (1).

C4. Guard cells control the opening and closing of stomata (1); e.g. opening them for gas exchange, or closing them to reduce water loss (1).

Section D — Factors affecting transpiration rate

D1. Any four: temperature (1); humidity (1); air movement/wind (1); light intensity (1).

D2. Higher temperature gives water molecules more energy, so they evaporate faster (1); this increases the rate of diffusion out of the stomata (1).

D3. Wind removes water vapour from around the leaf (1), maintaining a steeper concentration gradient between the inside and outside of the leaf, increasing diffusion (1).

D4. A potometer (1).

Marking note: accept any scientifically correct alternative answer with the same reasoning, even if different from the example shown — award marks for a correctly linked point, not for matching these exact words.